Ajan Muthuraj

Robotics Software & Autonomy

Japan | ajanm2003@gmail.com | Portfolio | GitHub | LinkedIn

Mechanical engineering graduate from IIT Madras and Systems Engineer in Japan. Built ROS 2 systems for navigation, perception and manipulation, with hands-on UR5/eBot integration and research in multi-robot autonomy and reinforcement learning.

EXPERIENCE

Systems Engineer, Dai-ichi Life Techno Cross (DLTX) | Japan | Oct 2026 - Present

- Practiced reverse and forward engineering of Lotus Notes applications. Built a React interface from supplied XML exports.
- Built foundations in full-stack development, including Java/Spring Boot backends, data persistence, Docker and Kubernetes fundamentals.

Robotics Research Intern, IIT Roorkee | India | Jun - Jul 2025

- Integrated Nav2, SLAM and RGB-D object localization in a ROS 2/Gazebo hospital humanoid built from SolidWorks-derived URDF. Used MoveIt 2 and OctoMap for collision-aware grasps, with about 96.6% success across 30 reported simulation trials.

SELECTED ROBOTICS PROJECTS

Isaac Sim UAV Autonomy | Integrated ROS 2, ArduPilot SITL and RTAB-Map with collision-aware 3D frontier exploration. Published container definitions, launch instructions and a nine-drone simulation.

e-Yantra Logistic coBot | Integrated a UR5 and eBot using MoveIt 2 Servo, ArUco pose estimates, EKF localization and ultrasonic docking. Tested in Gazebo and on hardware at IIT Bombay.

Time-Varying Reach-Avoid RL Thesis | Trained a DDQN on custom environments with moving targets and obstacles. Achieved 79.5% and 76.0% grid rollout success in two simulations and 94.4% predicted-safe precision in the two-obstacle setup.

Swarm Rescue Challenge | Contributed to FastSLAM/ICP localization and collaborative exploration for a simulated fleet of 10 UAVs. Our IIT Madras team placed second among Indian teams. Related work was presented as an ICRA 2026 XPLORE Workshop poster.

Adaptive RRT* and LPA* | Compared replanning algorithms in dynamic scenarios using path length, robustness and computation as evaluation criteria.

GeoGuide Overhead-Vision Rover | Combined ResNet-50 event detection and Dijkstra routing with ArUco tracking and an ESP32 rover receiving commands over Wi-Fi/TCP.

TECHNICAL SKILLS

Software: Python, C++, Linux, Git, ROS 2, Gazebo, Isaac Sim, Nav2, MoveIt 2, OpenCV, PyTorch, MATLAB.

Robotics and hardware: SLAM, sensor integration, motion planning, reinforcement learning, Arduino/ESP32, SolidWorks, Fusion 360.

EDUCATION AND LEADERSHIP

IIT Madras | B.Tech, Mechanical Engineering | 2022 - 2026 | CGPA 8.52/10

Electronics Club and Electronics & Robotics Guild, IIT Madras: Led Cubitect prototyping, trained club members and organized teams for Inter-IIT robotics competitions.

Languages: English, Hindi, Tamil, Japanese (basic).